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NETracer: A Topology-Aware Iterative Tracing Approach for Tubular Structure Extraction

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Abstract

Extracting tubular structures from images is a widespread and challenging task in computer vision. To explore these continuous structures, iterative tracing methods offer a promising direction. However, in scenes with dense and blurred branches, existing tracing methods tend to jump to adjacent branches during the tracing process, leading to a significant topological mistake. The reason for this shortcoming is that the tracing model only focuses on the estimation of discrete nodes and ignores their connection attribution. To solve this problem, we introduce NETracer, a topology-aware iterative tracing method to improve the continuity and topological accuracy. In our approach, a node-edge estimation network with local connectivity loss is trained to produce the future nodes and their connective edges. Then, a geodesic distance-based search strategy is employed with the help of predicted edge cues to trace the future branches more accurately. Additionally, to comprehensively assess the effect of the tracing model, a new tracing metric is proposed to evaluate the local accuracy, continuity, and topological correctness of the traced branches. We demonstrate that our method outperforms existing segmentation and tracing methods on five 2D road, vessel, and 3D neuron datasets. The code of NETracer is available at https://github.com/CSDLLab/NETracer.

1. Introduction

Tubular structure extraction is a crucial and challenging task in computer vision, such as the reconstruction of neuronal dendrites or axons [15, 24], the segmentation of retinal vessels [28], and the delineation of road networks from aerial imagery [5, 33]. These tubular structure can be represented as a series of nodes and edges connecting them, and our objective is to precisely extract these continuous objects to align with the image evidence.

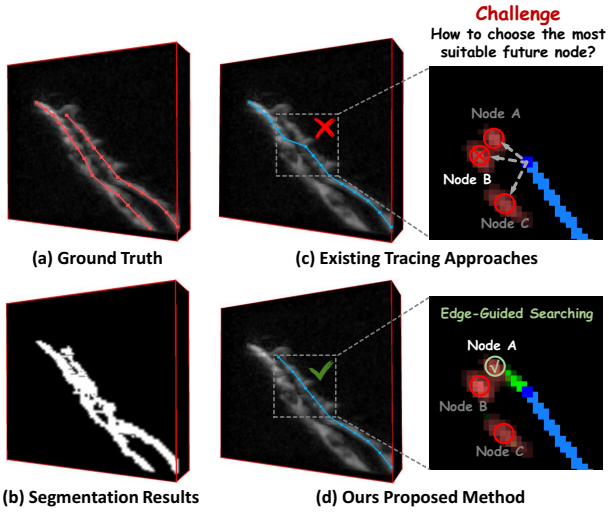


Figure 1. Motivation. (a) 3D neuron image and ground truth. (b) Segmentation results misinterpret ambiguous branches as a single entity. (c) Existing tracing approaches overlook the connectivity between discrete nodes, resulting in topological errors. (d) Integrating edge information can improve the topological accuracy.

Most existing methods tackle this problem as a **two-stage segmentation task**: first, employing a segmentation network [13, 25, 27, 44] to learn the centerline features, followed by the use of a post-processing algorithm [11, 12, 19, 32] to extract the topological connectivity patterns. However, the major drawback of these approaches is that they place significant reliance on the segmentation results from the first stage. When the signal-to-noise ratio (SNR) of image evidence is low, segmentation results may exhibit merging or blurring in localized regions, as illus-

trated in Fig. 1 (b). This issue can mislead post-processing algorithms into producing *shortcut* topological errors.

In contrast, **iterative tracing approaches** provide a more streamlined method to explicitly model the continuity of the branches. They start from a set of seed nodes and progressively explore unknown branches. The key to these methods lies in estimating the position of next node to drive the iterative process. This encompasses estimating the branch direction [1, 4, 14] and deriving the future node position from a probability map [10, 30]. However, the shortcoming of these approaches is that they treat the extraction of continuous curves as a discrete node prediction task, lacking the ability to learn the connectivity features between adjacent nodes. When encountering scenarios where two branches are closely spaced or entangled, multiple potential future nodes may become available for selection, as shown in Fig. 1 (c). Thus, a natural question will be raised: *How to choose the most suitable future node?*

To address this challenge, we introduce NETracer, a novel topology-aware iterative tracing model. Our proposed method aims to model the tubular structure as continuous sub-segments rather than discrete nodes, thereby integrating the local connectivity patterns of these nodes into the tracing process. Firstly, we theoretically analyze the limitations of existing tracing methods, and propose a new optimization objective that emphasizes branch continuity. Following this criterion, a node-edge estimation network (NENet) is proposed which includes three subtask learning blocks: centerline, future node, and future edge. The additional future edge information can provide crucial connectivity cues for guiding the selection of future nodes. Notably, a local connectivity loss is added in the future edge estimation block to ensure the reliability of local edge prediction. In the tracing process, a geodesic distance-based search strategy (GDS) is developed to select the most suitable future node from several potential node sets with the predicted edge cues. Finally, since there are few evaluation metrics for tracing models, we propose an iterative tracing metric that encompasses three sub-metrics to evaluate the continuity and topological accuracy of the tracing results. Our contributions are summarized as follows:

- We introduce NETracer, a novel topology-aware iterative tracing model. It seamlessly integrates connectivity features into the tracing process for the first time.
- NETracer employs a NENet with a local connectivity loss to produce future nodes and edges. Then, a GDS strategy is applied to pinpoint the most suitable future node based on the predicted edge cues.
- A brand-new iterative tracing metric is introduced to assess the local position accuracy, continuity, and topological accuracy of the tracing model.
- The extensive experiments on five 2D and 3D datasets show that our NETracer model surpasses other state-of-the-art (SOTA) segmentation and tracing methods.

2. Related work

2.1. Two-stage Segmentation Approaches

As mentioned in the previous section, these two-stage methods divide the feature learning of tubular structure into two distinct phases. Firstly, a segmentation network is trained to learn the centerline features. Then, a post-processing algorithm is applied to extract their topological connectivity patterns. In recent years, various centerline segmentation networks have demonstrated significant progress in addressing complex tubular structure extraction tasks and can be categorized into two primary groups. One category focuses on enhancing the network backbone, with notable contributions from the incorporation of attention mechanisms [20], deformable convolutions [6, 7, 25], and graph convolutions [41, 43], among others. Other approaches focus on refining downstream tasks or imposing specialized loss functions, including the topological constraint loss [27, 37], regional connectivity [21, 31], and tubular flux features learning block [34].

However, segmentation results can only provide a binary mask of the tubular structure, which is insufficient for constructing the final representation. To address this, various post-processing techniques have been developed. APP and APP2 [23, 38] apply a fast marching method using geodesic distance to reconstruct grayscale neuron images. Path-CNN [11] employs a progressive minimal path method on the segmentation centerline, providing geometric cues to improve performance. These methods directly extract curves from a global predicted centerline map without imposing any topological constraints, resulting in low reliability of the local structures. As an improved method, [19] extracts all significant nodes and applies the A* algorithm to connect them into an overcomplete graph. Subsequently, a path classifier is trained to determine whether the given path exists. However, this method still faces the same challenge: its path classifier must be an extremely robust binary classification model. Otherwise, the probability of generating shortcut branches rises significantly.

2.2. Iterative Tracing Approaches

Iterative tracing approaches aim to estimate the position of future nodes to finish the tracing process. To capture the continuous features of branches, some methods rely on parametric geometric models designed based on prior knowledge. neuTu [42] utilizes a cylinder template to fit the branch in the image. [17, 36] employs a ray burst sampling method to recursively explore along the neurite branches. [26] proposes the voxel-scooping spherical model to iteratively carve out the neuron branch. However, these methods are only effective in certain applications and rely on manu-

ally crafted features, making it difficult to tune parameters when dealing with different datasets.

With the advent of deep learning networks, RoadTracer [1] and SPE [4] construct the CNN-based models to learn the direction and radius of branches to estimate the next iteration step. However, the estimation of these high-level semantic features of branches consistently leads to positional errors contrasted with the ground truth, and this deviation tends to increase throughout the iterative tracing process. To enhance the accuracy of the tracing process, DeepBranchTracer [14] takes global centerline features into consideration to adjust the position of predicted future nodes at each step, effectively increasing the continuity of the branches. As another simplified solution, Vec-Road [30] proposes that future nodes can be obtained directly from the peak of the predicted position distribution probability maps. Nonetheless, all of these tracing approaches do not consider the connectivity feature of two neighboring nodes, and the predicted nodes may jump to other adjacent branches, leading to significant topological errors.

3. Method

In this section, we formulate the tubular structure tracing problem as an optimization objective, encompassing the posterior probability of historical paths and the continuity of branches. Based on this principle, we propose the NENet to learn the local branch attributions, which integrate the centerline, future node, and future edge blocks with a topology-aware constraint loss. In the tracing process, a geodesic distance-based search strategy is proposed and can search along the edge probability map to candidate the most suitable future node from the several potential future nodes.

3.1. Problem Formulation

A tubular structure is a continuous curve traversing 3D space, formally denoted as $\{\mathbf{c}(t) = (x(t), y(t), z(t))^T, t \in (0, 1)\} \in C$, where $C \subset \mathbb{R}^3$. The extraction of tubular structure is trying to seek the optimal curve $\mathbf{c}$ given the image evidence I, and can be represented as $P(\mathbf{c}|I)$. For the tracing models, they discretized this curve into M discrete nodes $\mathbf{c}(t_i)$ and the tracing task can be formulated within a probabilistic framework:

$$P(\mathbf{c}|I) \propto P(\mathbf{c}(t_1)|I) \prod_{i=2}^{M} P\left(\mathbf{c}(t_i)|W[\mathbf{c}(t_1), \cdots, \mathbf{c}(t_{i-1})], I\right), \tag{1}$$

where I is the input image. The first term $P(\mathbf{c}(t_1)|I)$ represents the initial seed node for the tracing process, and the second term can be represented as estimating the future nodes by maximizing a posterior probability given the image evidence I and the historical path W. Thus, the optimization objective for iterative tracing approaches can be written as:

$$c^* = \arg\max_{\mathbf{c}(t_i) \in C} P(\mathbf{c}(t_i)|W[\mathbf{c}(t_1), \cdots, \mathbf{c}(t_{i-1})], I). \tag{2}$$

However, as an approximate optimization objective, the continuity and connectivity between two neighboring nodes $\mathbf{c}(t_i)$ and $\mathbf{c}(t_{i-1})$ is ignored, leading to critical topological errors in the tracing scenario of two adjacent branches, as shown in Fig. 1(c). To tackle this problem, a better tracing model should estimate not just the future nodes but also the edges that interconnect them, aiming to enhance the topological integrity. Building upon this concept, a new optimization objective can be formalized as follows:

$$c^* = \arg\max_{\mathbf{c}(t_i) \in C} \int_{t_{i-1}}^{t_i} P(\mathbf{c}(t)|W[\mathbf{c}(t_1), \cdots, \mathbf{c}(t_{i-1})], I) \mathrm{d}t, \tag{3}$$

where the integral from t_{i-1} to t_i on the curve can capture the probability of the continuous edge from the current node $\mathbf{c}(t_{i-1})$ to the future node $\mathbf{c}(t_i)$. Based on this criterion, the model can learn to predict future nodes that not only individually align well with the image evidence, but also form a sequence that adheres to the expected topology.

3.2. NENet Architecture

Following the above optimization objective, the design of a deep network for iterative tracing can be decoupled into three subtasks:

- *Exploring Future Nodes*: The primary task of an iterative tracing model is to explore future nodes along the branch.
- *Stopping Criteria*: The iterative process needs stopping criteria, which determine when there are no more nodes to explore, or the current node is far from the centerline.
- *Providing Seed Nodes*: A robustness seed generator is essential for initiating the tracing process.

Here, we introduce the node-edge estimation network (NENet). The NENet contains three sub-task learning blocks: the centerline segmentation block, the future node estimation block, and the future edge estimation block, as depicted in Fig. 2.

Global Centerline Segmentation. Minimizing the centerline segmentation loss can be viewed as a variant optimization objective $P(\mathbf{c}|I) \propto \prod_{i=1}^{m} P(\mathbf{c}(t_i)|I)$, emphasizing the fit of each node position to the image data. Although there is no topological relationship between these nodes, they can still provide a robust likelihood estimate for determining whether a given position lies on the curve. Thus, the centerline segmentation results can produce the *off-centerline stopping criteria* and the initial *seed nodes* based on a simple threshold method for the tracing process without additional specially designed seed generation modules.

In our network architecture design, we adopt the V-Net architecture [16] as the backbone of the model, chosen for

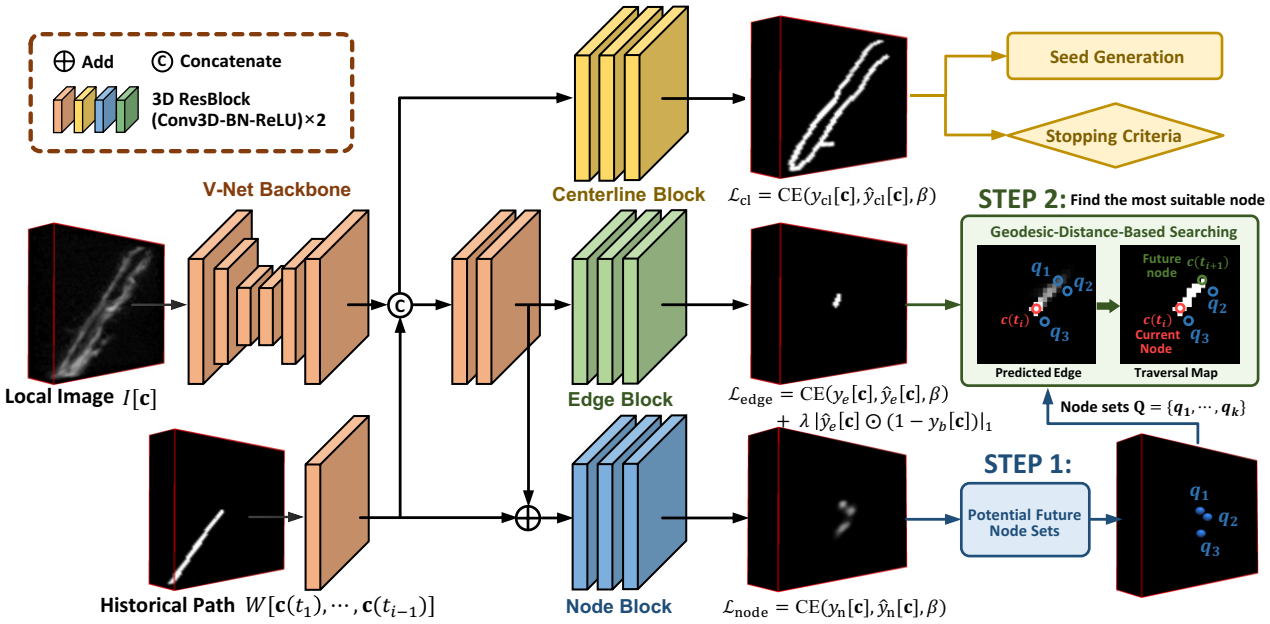


Figure 2. Overview of NETracer Architecture. Our network contains three sub-task learning blocks. Combined with the continuity prior from the centerline learning block, future node and edge estimation tasks can achieved better performance. In the tracing process, the node block predicts the future node sets $\mathbf{Q}$. Then, a GDS strategy is utilized with the predicted edge cues to find the most suitable future node.

its lightweight design and robustness. The weighted Cross-Entropy loss [39] is employed to optimize the centerline segmentation task while imposing a higher cost $\beta = 0.9$ for the foreground branches:

$$\mathcal{L}_{\text{cl}} = \text{CE}(y_{\text{cl}}[\mathbf{c}(t_i)], \hat{y}_{\text{cl}}[\mathbf{c}(t_i)], \beta), \tag{4}$$

where $y_{\text{cl}}[\mathbf{c}(t_i)]$ and $\hat{y}_{\text{cl}}[\mathbf{c}(t_i)]$ represent the binary centerline label and predicted probability map at the position of node $\mathbf{c}(t_i)$. The seed nodes are randomly selected on the predicted centerline, and the threshold for off-centerline stopping criteria is set to 0.5.

Future Node Block. In previous work [30], the future node estimation block is trained to produce the potential successor nodes based on the location of the maximum value in the probability map, and the loss function can be formulated as:

$$\mathcal{L}_{\text{node}} = \text{CE}(y_n[\mathbf{c}(t_i)], \hat{y}_n[\mathbf{c}(t_i)], \beta), \tag{5}$$

where $y_n[\mathbf{c}(t_i)]$ is the binary label map of future node at $\mathbf{c}(t_i)$. Specifically, the future node label $y_n[\mathbf{c}(t_i)]$ assigns 1 to pixels $\mathbf{p}$ located within a radius r around the next node $\mathbf{c}(t_{i+1})$, and 0 for the background:

$$y_n[\mathbf{c}(t_i)] = \begin{cases} 1, & \text{if } \|\mathbf{p} - \mathbf{c}(t_{i+1})\|_2 \le r \\ 0, & \text{otherwise} \end{cases}. \tag{6}$$

However, when confronted with adjacent or tangled branches, there will be several peaks in the future node probability map, and how to locate the most suitable node becomes an interesting problem.

Local Future Edge Block. Drawing from the optimization criterion outlined in Eq. (3), we propose a future edge estimation block to provide the local connectivity cues for the future node selection. The loss function is formulated as:

$$\mathcal{L}_e = \text{CE}(y_e[\mathbf{c}(t_i)], \hat{y}_e[\mathbf{c}(t_i)], \beta), \tag{7}$$

where $y_e[\mathbf{c}(t_i)]$ is the binary label of the future edge at $\mathbf{c}(t_i)$. The label $y_e[\mathbf{c}(t_i)]$ assigns 1 for pixels $\mathbf{p}$ lying on the path from node $\mathbf{c}(t_i)$ to $\mathbf{c}(t_{i+1})$, and 0 for the background:

$$y_e[\mathbf{c}(t_i)] = \begin{cases} 1, & \text{if } \mathbf{p} \in \mathbf{c}(t), t \in (t_i, t_{i+1}) \\ 0, & \text{otherwise} \end{cases}. \tag{8}$$

Although the CE loss and clDice loss are effective in capturing the global continuity of branches, they fall short in detecting local connectivity features when encountering adjacent branches. The edge block is expected to generate a conservative connection estimate, predicting only the local path from the current node to the correctly connected next node. To tackle this limitation, we incorporated a modified path label $y_b[\mathbf{c}(t_i)]$ with a local connectivity loss function $\mathcal{L}_{\text{conn}}$ to enhance the performance of edge block:

$$\mathcal{L}_{\text{conn}} = |\hat{y}_e[\mathbf{c}(t_i)] \odot (1 - y_b[\mathbf{c}(t_i)])|_1, \tag{9}$$

where $|\cdot|$ is L1 norm and $\odot$ denotes the element-wise multiplication. Unlike the centerline label $y_{\text{cl}}[\mathbf{c}(t_i)]$, $y_b[\mathbf{c}(t_i)]$

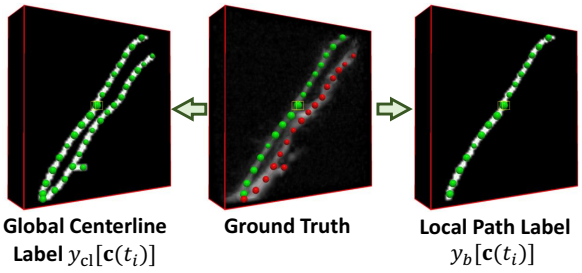


Figure 3. The difference between the centerline label $y_{\text{cl}}[\mathbf{c}(t_i)]$ and the local path label $y_b[\mathbf{c}(t_i)]$. y_b contains only the local path connected to the current node $\mathbf{c}(t_i)$, rather than all the branches.

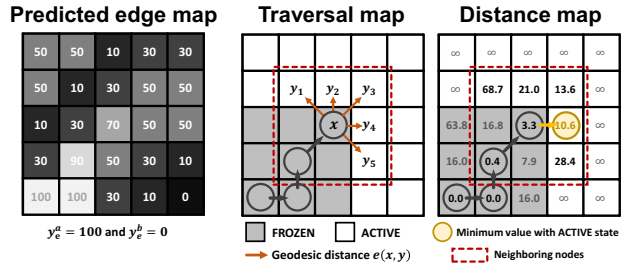


Figure 4. The illustration of geodesic distance-based search strategy. Specifically, we employ 8-neighborhood connectivity for 2D images and 26-neighborhood connectivity for 3D images.

contains only the local branches connected to the current node, rather than all the branches, as shown in Fig. 3. The local connectivity loss $\mathcal{L}_{\text{conn}}$ can penalize any predicted edges deviating from the current branches. When y_b is 1, indicating the correct connection patterns and the loss is 0. If y_b is 0, the loss will becomes $|\hat{y}_e|_1$, penalizing incorrect edge predictions in background regions. With the help of $\mathcal{L}_{\text{conn}}$, the edge block will be more sensitive to local connectivity. Thus, the final loss for edge block is:

$$\mathcal{L}_{\text{edge}} = \mathcal{L}_e + \lambda \mathcal{L}_{\text{conn}}. \tag{10}$$

Here, λ is set to 1 to balance the two loss functions.

Additionally, NENet is an end-to-end network of multi-task joint learning. The centerline features and historical traced path can provide a robust continuity prior for the learning of subsequent nodes and edges. Thus, the concatenation of these two features is set as the input of future node and edge estimation blocks. The overall losses of NENet are integrated as:

$$\mathcal{L} = \mathcal{L}_{\text{cl}} + \mathcal{L}_{\text{node}} + \mathcal{L}_{\text{edge}}. \tag{11}$$

3.3. Geodesic Distance-based Search Strategy

In the tracing process, as previously discussed, a significant challenge lies in selecting the most suitable node from multiple potential future nodes. To solve this problem, we propose a geodesic distance-based search strategy, which initiates from node $\hat{\mathbf{c}}(t_i)$ and follows the predicted edge cues $\hat{y}_e[\mathbf{c}(t_i)]$ to identify the future node $\hat{\mathbf{c}}(t_{i+1})$ with topological reliability.

Potential Future Node Sets. To extract all potential future nodes from the probability map $\hat{y}_n[\mathbf{c}(t_i)]$, we initially locate the maximum pixel/voxel in this probability map and add it into the future node set $\mathbf{Q}$. Then, the surrounding area of this node will be masked with 0. Subsequently, we attempt to find a new maximum pixel/voxel in this masked probability map. When the maximum value in the masked probability map decays to half of the original maximum value, it indicates that most of the potential nodes have been identified, and the node selection process is terminated.

Geodesic Distance Searching. After obtaining the potential node sets $\mathbf{Q}$, we start from the current node $\mathbf{c}(t_i)$ and trace along the predicted edge $\hat{y}_e[\mathbf{c}(t_i)]$ with a geodesic distance to find the most suitable future node $\hat{\mathbf{c}}(t_{i+1})$. Initially, two distance map and traversal map are constructed, which have the same dimensions as the edge image. The distance map stores the geodesic distance from the current node $\mathbf{c}(t_i)$ to other nodes in the map, and is initialized with ∞. The traversal map stores the state of whether each location has been traversed or not, and is initialized with *ACTIVE* state.

Starting from the position $\mathbf{x}$, its neighboring nodes $\mathbf{y}$ in distance map are updated as follows: $d(\mathbf{y}) = \min\{d(\mathbf{y}), d(\mathbf{x}) + e(\mathbf{x}, \mathbf{y})\}$. Here, $d(\mathbf{x})$ represents the distance value at position $\mathbf{x}$, and $e(\mathbf{x}, \mathbf{y})$ is the geodesic distance between these two neighboring nodes, calculated as follows:

$$e(\mathbf{x}, \mathbf{y}) = ||\mathbf{x} - \mathbf{y}|| \cdot \frac{g(\mathbf{x}) + g(\mathbf{y})}{2}, \tag{12}$$

where $||\mathbf{x} - \mathbf{y}||$ indicates the Euclidean distance, and $g(\cdot)$ evaluates the difference between the current nodes and foreground:

$$g(\mathbf{x}) = \exp\left(\gamma \cdot \frac{\hat{y}_e^a - \hat{y}_e(\mathbf{x})}{\hat{y}_e^a - \hat{y}_e^b}\right) - 1, \tag{13}$$

where $\hat{y}_e^a$ and $\hat{y}_e^b$ are the maximum and minimum values of the edge probability map, and $\gamma = 5$ is a scaling coefficient. After updating the distance value of the neighboring nodes $\mathbf{y}$, the next node to be traversed is the minimum value in distance map with *ACTIVE* state. Then, the state of these neighboring nodes in traversal map will change to *FROZEN* state. This process continues until the first node $\hat{\mathbf{c}}(t_k)$ in the potential future node set $\mathbf{Q}$ is found. The details of the GDS strategy is shown in Fig. 4.

3.4. Training Details

Training Samples with Path Correction. A common challenge faced by tracing methods is that there will be slight

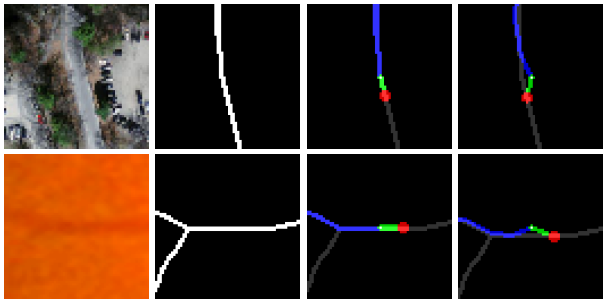


Figure 5. Illustrations of training samples with path correction from ROAD and DRIVE datasets. The historical path, future edge, and future node are shown in blue, green and red, respectively.

position error for future nodes estimation, and it will accumulate over the tracing process until the trajectory deviate from the centerline. To solve this problem, we conduct the training samples with path correction, hoping to improve the position accuracy of the tracing results. In the previous method, the training samples are derived from the centerline curve. In this paper, to enhance the path correction ability of the model, a slight noise is injected into the position of historical path and current node, as shown in Fig. 5. Specifically, the movement of these noise control nodes is confined to the plane perpendicular to the current centerline. The size of the training samples are 64×64 for 2D images, and $32 \times 64 \times 64$ for 3D images.

Training Details. The network architecture is implemented in PyTorch 1.13.0 and MindSpore 1.7.0. The hardware configuration includes an Intel Xeon Platinum 8383C CPU, and two NVIDIA GeForce RTX3090 GPUs. With this server configuration, we took about 8 hours to train the model.

4. Experiments

4.1. Datasets and Evaluation Metrics

Datasets. We conduct experiments on five popular datasets containing 2D and 3D images: Massachusetts Roads Dataset [18], DRIVE and CHASE_DB1 vessel dataset [3, 29], fMOST-VTA and DIADEM-CCF neuron dataset [2, 8, 9]. Further details about the datasets are provided in the Appendix.

Global Coverage Metric. Two metrics are used to evaluate the integrality of tracing results. SSD-F1 metric [22] measures the number of nodes matched between the test and gold branches. Length metric (L-F1) [35, 40] is a variant of SSD-F1 that measures the number of matched edges.

Iterative Tracing Metric. Current evaluation metrics always focus on the coverage of the centerline and lack an analysis of the topological correctness of the traced branches. Thus, we propose the brand-new iterative tracing metrics, illustrated in Fig. 6. The Position Error (PE) measures the Euclidean distances between the positions of matching nodes in the test and gold branches, aiming to evaluate the local accuracy of the traced branches. The Average Branch Length (ABL) is designed to evaluate the continuity of branches, hoping the results contain longer and more complete branches. Another common mistake occurs when two branches are close together, causing the tracing models jump to the wrong branch. Therefore, we introduce the Jump Error (JE) to evaluate these critical topological errors. Specifically, for each node $\mathbf{n}_t$ and its parent node $\rho(\mathbf{n}_t)$ on the test graph, their matching nodes $\mathbf{n}_g^*$ and $\mathbf{n}_{g_p}^*$ can be found on the ground truth graph. If the graph distance between these two nodes is greater than their straight-line distance, a jump error is considered to have occurred. More details for PE and JE metrics are laid out in Appendix.

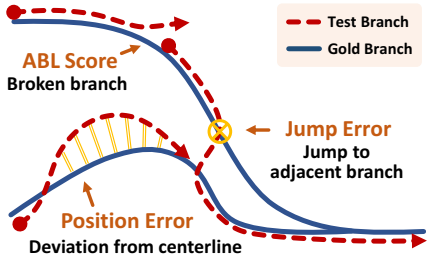


Figure 6. Visualization of three iterative tracing metrics, including ABL score, PE score and JE score.

4.2. Comparison with SOTA Methods

We compare our proposed model with several SOTA segmentation and tracing methods. For segmentation models, we select V-Net [16], CS-Net [20], clDice [27] and DSCNet [25] as baselines. The output of the centerline block in our model is denoted as NENet-Seg. For iterative tracing methods, we choose neuTu [42], RoadTracer [1], Vec-Road [30], DeepBranchTracer (DBT) [14], and SPE [4] as comparison.

Road Dataset. The tracing results of the Massachusetts Roads Dataset are shown in the first row of Fig. 7. In scenarios involving the tracing of multiple adjacent roads, the NETracer consistently delineates the road topological structures without abrupt jumps between adjacent roads. The quantitative analysis of the model is shown in Table 1. It is notable that the state-of-the-art segmentation methods like clDice and DSCNet exhibit performance comparable to our method in the coverage metric. However, their topological accuracy, assessed by ABL, PE, and JE, is markedly inferior to our proposed method, particularly on the JE metric. This is mainly because their segmentation mask fails to provide sufficient explicit topological cues for the post-processing skeletonization algorithm. In contrast, iterative tracing approaches, such as Vec-Road, have significant advantages in tracing metrics. Our proposed method achieves the high-

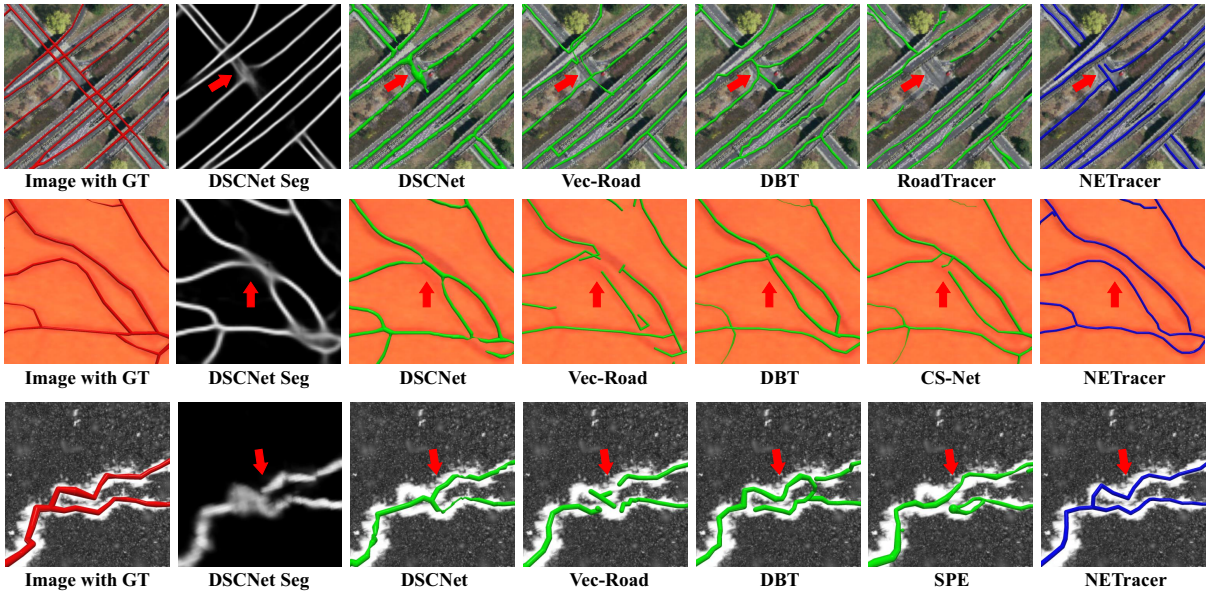


Figure 7. Tracing results of 2D road, vessel, and 3D neuron datasets. The red arrow indicates that our model can achieve the best topological accuracy and continuity when confronted dense or adjacent branches in the image.

	Model	Coverage Metric		Tracing Metric		
		SSD-F1	L-F1	PE ↓	JE ↓	ABL ↑
ROAD	V-Net†	0.690	0.803	1.499	19.3	156.0
	V-Net+clDice †	0.711	0.815	1.321	12.9	169.3
	DSCNet †	0.713	0.828	1.404	20.7	131.3
	RoadTracer	0.289	0.552	3.220	20.8	66.1
	Vec-Road	0.626	0.794	1.623	13.4	106.8
	DBT	0.680	0.816	1.372	24.3	200.2
	NENet-Seg†	0.683	0.803	1.511	17.5	154.4
	NETracer	**0.718**	**0.836**	**1.263**	**8.2**	**234.4**
DRIVE	V-Net†	0.898	0.817	0.643	2.5	44.5
	V-Net+clDice †	0.898	0.786	0.675	3.1	53.9
	DSCNet †	0.919	0.820	0.687	2.2	56.9
	CS-Net†	0.899	0.821	0.647	2.3	44.0
	Vec-Road	0.901	0.763	0.696	5.5	40.9
	DBT	0.909	0.822	0.715	6.2	67.7
	NENet-Seg†	0.900	0.811	0.675	2.4	45.7
	NETracer	**0.925**	**0.824**	**0.633**	**1.9**	**74.4**
CHASE_DB1	V-Net†	0.765	0.825	**0.763**	7.8	107.8
	V-Net+clDice †	0.825	0.838	0.912	8.2	108.7
	DSCNet †	0.783	0.807	0.887	6.3	100.7
	CS-Net†	0.766	0.828	0.773	7.5	128.5
	Vec-Road	0.821	0.842	0.852	4.9	122.6
	DBT	0.833	0.843	1.034	8.8	124.7
	NENet-Seg†	0.771	0.819	0.871	6.6	110.8
	NETracer	**0.837**	**0.860**	0.771	**4.3**	**131.6**

Table 1. Quantitative evaluation for 2D road and vessel datasets. † means that these are segmentation methods and employ neuTu [42] as post-processing approach to produce their final output.

est coverage score while maintaining satisfactory continuity and topological accuracy across all metrics.

Vessel Dataset. Most existing methods consider the DRIVE and CHASE_DB1 vessel dataset as segmentation problems. When two vessels are in close proximity, these methods often fail to distinguish them accurately, as highlighted by the red arrow in the second row of Fig. 7. Our NETracer model stands out as the only method to trace the correct topology accurately. The quantitative evaluation of tracing results is shown in Table 1.

	Model	Coverage Metric		Tracing Metric		
		SSD-F1	L-F1	PE ↓	JE ↓	ABL ↑
fMOST-VTA	V-Net †	0.854	0.854	0.700	4.5	125.6
	V-Net+clDice †	0.907	0.898	0.695	3.3	150.8
	DSCNet †	0.902	0.895	0.700	3.1	120.0
	neuTu	0.691	0.689	1.089	3.3	56.2
	SPE	0.807	0.860	1.201	4.8	94.9
	Vec-Road	0.901	0.898	0.706	3.5	151.1
	DBT	0.897	0.900	0.941	4.4	167.9
	NENet-Seg†	0.851	0.853	0.833	9.6	140.6
	NETracer	**0.919**	**0.913**	**0.693**	**2.3**	**219.3**
DIADEM-CCF	V-Net †	0.776	0.738	2.540	5.7	84.9
	V-Net+clDice †	0.737	0.698	2.583	9.1	60.3
	DSCNet †	0.701	0.667	2.554	8.3	42.6
	SPE	0.772	0.723	2.507	7.6	69.4
	neuTu	0.401	0.527	2.646	5.3	49.4
	Vec-Road	0.763	0.731	2.511	6.4	100.7
	DBT	0.783	0.753	**2.422**	10.3	95.7
	NENet-Seg†	0.733	0.690	2.539	7.1	75.2
	NETracer	**0.792**	**0.755**	2.491	**5.1**	**107.1**

Table 2. Quantitative evaluation for 3D neuron datasets. † means that these are segmentation-based methods.

Neuron Dataset. We also test our model on 3D fMOST-VTA and DIADEM-CCF neuron datasets, which are more

Model	Ablation of Each Module				Coverage Metric		Tracing Metric		
	$\mathcal{L}_{\text{node}}$	$\mathcal{L}_e$	$\mathcal{L}_{\text{conn}}$	GDS	SSD-F1	L-F1	PE ↓	JE ↓	ABL ↑
M1	✓				0.892	0.888	0.746	36	142.8
M2	✓	✓			0.897	0.895	0.734	32	171.7
M3	✓			✓	0.893	0.887	0.750	51	158.8
M4	✓	✓	✓		0.903	0.901	0.702	26	185.6
M5	✓	✓	✓	✓	**0.919**	**0.913**	**0.693**	**18**	**219.3**

Table 3. Ablation studies of our method on 3D fMOST-VTA dataset. Note that all the experiments use the same seed nodes to finish the tracing process.

Model	Coverage Metric		Tracing Metric			Runtime		
	SSD-F1	L-F1	PE ↓	JE ↓	ABL ↑	Net Inf.	Tracing	Total
SPE	0.807	0.860	1.201	4.8	94.9	56.3s	18.9s	75.2s
Vec-Road	0.901	0.898	0.706	3.5	151.1	71.2s	13.3s	84.5s
DBT	0.897	0.900	0.941	4.4	167.9	260.9s	17.5s	278.4s
NETracer	**0.919**	**0.913**	**0.693**	**2.3**	**219.3**	72.4s	25.4s	97.8s

Table 4. Time consumption analysis of SOTA iterative tracing methods on 3D fMOST-VTA dataset.

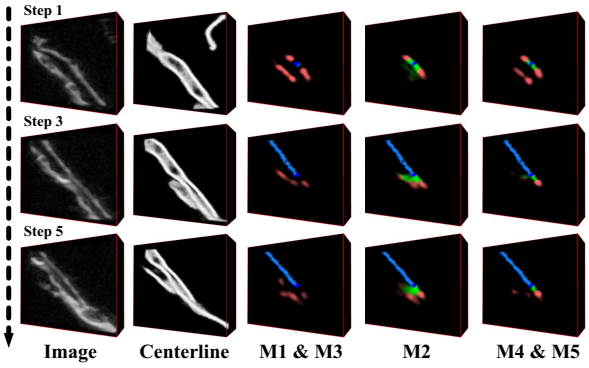


Figure 8. Visualization of the predicted future node and edge probability maps in the tracing process. The future nodes (in red), future edges (in green), and the historical path (in blue) are visualized by overlaying in the same image.

challenging than the 2D images due to increased image dimensions. As depicted in the third row of Fig. 7, even when confronted with low confidence in centerline segmentation outcomes, our method consistently preserves the accuracy of the topological structure. Quantitative evaluation in Table 2 confirms the effectiveness of NETracer.

Time Consumption. We conduct the time consumption analysis of our model on 3D neuron datasets, as shown in Table 4. Three tracing models are chosen for comparison and the time consumed at the stages of network inference and tracing is reported. Our proposed model contains a GDS module which will slightly increase the time consumption at the stage of tracing, but can achieve significantly improved PE, ABL and JE scores compared to other methods.

4.3. Ablation Study

We conduct ablation studies to demonstrate the improvement conferred by the integration of each module within our NETracer method, as shown in Table 3. Additionally, we visualize the tracing process of the model to show the effects of each module of the model.

Limitation of Existing Iterative Tracing Method. M1 only utilizes predicted nodes feature to trace the branches. When confronted with two adjacent branches, there will be multiple potential future nodes to be selected, as shown in the second column of Fig. 8. Relying solely on the maximum of probability map to identify the future node can lead to topological errors, resulting in the worst JE score.

Local Connectivity Loss. M2 leverages edge information to enhance the prediction of future nodes, and achieves a slight improvements on both coverage and tracing metrics. However, a network trained solely on the CE edge loss $\mathcal{L}_e$ will produces imprecise edge prediction and is inadequate to capture the continuous nature of branches, as observed in the third column of Fig. 8. To address this problem, M4 and M5 integrate the local connectivity loss $\mathcal{L}_{\text{conn}}$ to constrain the edge connection and achieve better edge accuracy.

Improvement by GDS Strategy. M5 utilizes the GDS strategy to find the future node. The highest SSD-F1, L-F1, and ABL scores, as well as the lowest PE and JE scores, indicate that this combination of techniques enhances the continuity and topological correctness of branches. Otherwise, to prove the necessity of the edge cues, we set M3 and apply the GDS strategy directly on the predicted centerline map. However, the performance of M3 is even worse than that of M1. This is due to the ambiguous centerline predictions misleading the GDS strategy to find the wrong future nodes. Therefore, the GDS strategy needs to combine with local connectivity loss to achieve optimal performance.

5. Conclusion

In this paper, we introduce NETracer, a novel topology-aware iterative tracing method that reliably identifies future nodes using predicted edge cues with the help of a geodesic distance-based search strategy. We also introduce three new metrics for iterative tracing that assess branch topological accuracy and continuity. Our method has been evaluated on five road, vessel, and neuron datasets, showing promising potential in extracting complex tubular structures.

Acknowledgements: This work was supported in part by the National Natural Science Foundation of China under Grant T2293723, in part by Zhejiang Provincial Natural Science Foundation under Grant LZ24F020003, in part by the Excellent Research Innovation Team of Anhui Provincial Education Department under Grant 2024AH010018, in part by the Leading Goose Research and Development Program of Zhejiang Province under Grant 2025C02045, and in part by Zhejiang Provincial Government (ZJU) Kunpeng and Ascend Center of Excellence.

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